Euler Methods

→ formula:

$$y_{n+1} = y_n + hf(x,y) \rightarrow \text{General sol}^n$$

→ Given that

$$\boxed{\frac{dy}{dx} = x+y}$$

Given that

$y(0) = 1$

$\boxed{x=0 \,/\, y=1}$ $[0,1]$

$h = \frac{1-0}{10} = \boxed{0.1}$

→ Formula: $y_{n+1} = y_n + hf(x,y) \rightarrow \text{General sol}^n$

$\boxed{n=0}$

$$y_1 = y_0 + hf(x,y)$$

$$y_1 = 1 + (0.1)(0+1)$$

$$\boxed{y_1 = 1.1}$$

$\boxed{n=1}$

$$y_2 = y_1 + hf(x_1, y_1)$$

$$y_2 = 1.1 + (0.1)(0.1+1.1) \quad \rightarrow h = x_1 + h = 0.1$$

$$y_2 = 1.1 + (0.1)(0.12) = 1.22$$

$\boxed{n=2}$

$$y_3 = 1.22 + (0.1)(0.2+1.22)$$

$$\boxed{= 1.362}$$

$\boxed{n=3}$

$$y_4 = 1.362 + (0.1)(0.3 + 1.362) = 1.5282$$

Ans

$$\Rightarrow \boxed{\frac{dy}{dx} = x+y}$$

by solving linear dE.eq.

$$\boxed{\frac{dy}{dx} - y = x}$$

$$P(x) = e^{-\int 1dx} \quad e^{-x} \quad \boxed{e^{-x}}$$

$$y \cdot e^{-x} = \int x \cdot e^{-x} \, dx.$$

BY using $\int u \cdot v$ we get.

$$\boxed{y = -x-1+2e^{x}}$$ Particular solution

Now,

$$y(0.1) = -0.1-1+2e^{0.1}$$
$$\boxed{= 1.110}$$

$$y(0.2) = -0.2-1+2e^{0.2}$$
$$= 1.243$$

$$y(0.3) - 0.3-1+2e^{0.3} =$$

Ans.

⇒

Calculating the error

$$E = \frac{Exact - Approx}{Ex} = \frac{(1.1) - (1.2)}{(1.1)} \approx 8\%$$